

AI for Financial Practitioners

Lecture 0 · Part I

One Financial Problem, Eight Generations of Intelligence

From relationships to organized judgment

The experiment: hold the problem constant

ONE FINANCIAL UNIVERSE · EIGHT ANALYTICAL LENSES

500 synthetic listed companies. One classification mandate.

- Observe growth, margins, ROIC, leverage, valuation, volatility, recurring revenue, cash conversion and momentum.
- Classify each company as **Defensive, Quality, Growth, Value, Cyclical, or Distressed**.
- Keep the economic world fixed; change only the **form of intelligence** applied to it.

Problem · classify investment profile

Evidence · same financial variables

Universe · same 500 companies

Only the intelligence changes

A

controlled experiment separates technological capability from changes in the task itself.

Eight generations — an expanding problem frontier

FROM MEASURABLE RELATIONSHIPS TO ORGANIZED JUDGMENT

01 · STATISTICS

Relationships + similarity

Question: What resembles what?

02 · ECONOMETRICS

Coefficients + probabilities

Question: What is conditionally likely?

03 · CLASSICAL ML

Nonlinear predictive rules

Question: What pattern predicts?

04 · UNSUPERVISED

Discovered structure

Question: What groups exist?

05 · NEURAL ML

Learned representations

Question: What internal features help?

06 · AUTOENCODER

Latent representation

Question: What coordinates can be learned?

07 · GENERATIVE AI

Meaning + explanation

Question: What does the evidence mean?

08 · AGENTIC AI

Organized judgment

Question: How should views be synthesized?

Generation 1–3: from resemblance to nonlinear prediction

STATISTICS · ECONOMETRICS · CLASSICAL MACHINE LEARNING

01 · STATISTICS

Mechanism

Standardize features;
compare distance to
profile centroids.

Adds

Relationships, similarity,
transparent baseline.

Boundary

No semantic or contextual
judgment.

02 · ECONOMETRICS

Mechanism

Multinomial logistic
regression.

Adds

Conditional probabilities
and interpretable
coefficients.

Boundary

Predefined variables,
labels and functional form.

03 · CLASSICAL ML

Mechanism

Random forest learns
nonlinear partitions.

Adds

Interactions and predictive
rules from examples.

Boundary

Still a supervised,
structured-data task.

Generation 4–6: the machine begins to learn structure

UNSUPERVISED LEARNING · NEURAL LEARNING · LATENT REPRESENTATION

04 · Unsupervised learning

K-means asks: *what structure exists without our labels?*



05 · Neural machine learning

Hidden layers learn nonlinear internal representations useful for classification.



06 · Autoencoder

A bottleneck learns compressed latent coordinates from the input structure itself.

The conceptual shift

- From reproducing a human taxonomy to **discovering structure**.
- From hand-specified features to **learned transformations**.
- From visible variables to **latent coordinates**.
- Human interpretation remains necessary: mathematical structure is not automatically economic meaning.

Generation 7: generative AI expands the representation of the problem

FROM A ROW OF NUMBERS TO AN ANALYST-STYLE DOSSIER

Structured evidence · growth · margin ·
ROIC · leverage · valuation · volatility



Natural-language dossier · financial facts
expressed in context



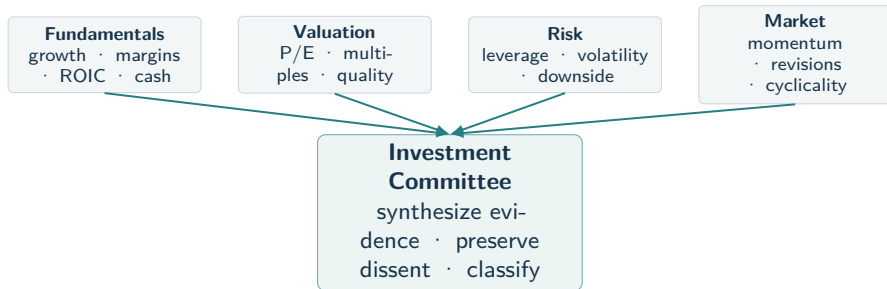
Claude Haiku 4.5 · classify · explain ·
identify evidence · state limitations

What changed?

- The model can operate on **language and meaning**.
- Evidence no longer has to fit a rigid tabular schema.
- The output can include an **explanation**, not only a class probability.
- New risks appear: variability, hallucination and overconfidence.

Generation 8: intelligence becomes organized judgment

SAME FOUNDATION MODEL · DIFFERENT INSTITUTIONAL ARCHITECTURE



What architecture adds

Independent perspectives · explicit roles ·
preserved disagreement · synthesis.

What architecture does not guarantee

More agents do not automatically create more
truth; coordination can amplify error or
groupthink.

Newer does not mean universally better

CHOOSE THE METHOD THAT FITS THE PROBLEM

Method	Especially strong when	Do not confuse it with
Statistics / econometrics	Structure is measurable; interpretation matters	Semantic understanding or autonomous judgment
Random forest	Stable labeled tabular prediction	A reason to replace every simpler model
Clustering	Categories are unknown or exploratory	Automatically meaningful economic segments
Neural / latent models	Useful representations must be learned	Transparent economic semantics
Generative AI	Evidence is unstructured and semantic	Exact deterministic computation
Agentic architecture	Multiple perspectives must be organized	Guaranteed correctness from more model calls

The comparative laboratory: hold the company constant

A STUDENT CAN SWITCH THE INTELLIGENCE — NOT THE PROBLEM

Select a company

Company-127

Select the generation

Statistics · Econometrics · Classical ML
· Unsupervised · Neural · Autoencoder
· Generative · Agentic

Statistics · resemblance

Econometrics / ML · probability + prediction

Unsupervised / neural · structure + representation

Generative AI · meaning + explanation

Agentic AI · deliberation + synthesis

The interface displays

- Output / classification
- What the generation contributes
- What it cannot do by itself
- The same underlying company dossier

From ladder to constellation

LECTURE 0 PART I ESTABLISHES THE VOCABULARY FOR INTELLIGENT SYSTEMS

01 · What **problem** are we trying to solve?

02 · What **information and representation** does it require?

03 · What form of intelligence is **best suited** to the task?

04 · What can the method **not** do reliably?

05 · How should multiple forms of intelligence be **combined and governed**?

The transition

Part I

The **ladder**: separate the generations to understand their capabilities.

Part II

The **constellation**: combine specialized forms of intelligence into one governed system.

Capability → architecture →
governed action